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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Machine Design) (CBCS) Degree Examinations

May - June 2024

22MMD201 – THEORY OF VIBRATIONS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Classify vibrations and discuss the effect of the mass of the spring on the natural frequency of a spring-mass system.	10	L*3	1	1
	b) What are beats? Derive a relation to determine their resultant amplitude.	7	L2	1	1
	c) An accelerometer mounted on a building frame indicates that the frame is vibrating harmonically at 15 cps, with a maximum acceleration of 0.5 g. Determine the amplitude and the maximum velocity of the building frame.	3	L3	1	1
2.	a) What is the use of logarithmic decrement? Derive an expression for the same.	7	L2	1	1
	b) For a vibrating system $m = 10$ kg, $k = 2500$ N/m and $c = 45$ N-s/m. A harmonic force of amplitude 180 N and frequency 3.5 Hz acts on the mass. If the initial displacement and velocity of the mass are 15 mm and 5 m/s, find the complete solution representing the motion of the mass.	13	L3	1	1
Unit – II					
3.	a) What are the different methods to control vibrations? With a neat sketch explain the active vibration isolation system.	8	L2	2	1
	b) A reciprocating engine has a mass of 40 kg and runs at a constant speed of 3000 rpm. After it was installed, it vibrated with a large amplitude at the operating speed, what dynamic vibration absorber should be coupled to the system if the nearest resonant frequency of the combined system must be at least 25 % away from the operating speed.	8	L3	2	1
	c) Explain the procedure of using Laplace transform with an example vibration problem	4	L2	2	1
4.	a) Derive an expression for the natural frequencies of an Undamped dynamic vibration absorber.	8	L3	2	1
	b) It is found that an exhaust fan of mass 80 kg and operating speed 1000 rpm, produces a repeating force of 10000N on its rigid base. If the maximum force transmitted to the base is to be limited to 2000 N using an undamped isolator, determine (i) the maximum permissible stiffness of the isolator that serves the purpose, (ii) the steady-state amplitude of the exhaust fan with the isolator that has the maximum permissible stiffness.	8	L3	2	1
	c) Derive the impulse response function.	4	L2	2	1
Unit – III					
5.	a) Discuss stability of equilibrium states.	5	L2	3	1
	b) Discuss iterative method to find the solution of an undamped vibrating system.	7	L2	3	1
	c) A spring-mass-damper system having an undamped natural frequency of 100 Hz and a damping constant of 20 N-s/m is used as an accelerometer to measure the vibration of a machine operating at a speed of 3000 rpm. If the actual acceleration is 10 m/s^2 and the recorded acceleration is 9 m/s^2 , find the mass and the spring constant of the accelerometer.	8	L3	3	2

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| 6. a) | What is phase plane representation? Give the phase plane representation of a simple harmonic oscillator. | 5 | L2 | 3 | 1 |
| b) | Discuss how a vibration pickup can be used as a 'vibrometer' and an 'accelerometer' and highlight the design considerations. | 6 | L3 | 3 | 1 |
| c) | A vibration pickup has a natural frequency of 5 Hz and a damping ratio of 0.5. Find the lowest frequency that can be measured with a 1 % error. | 5 | L3 | 3 | 1 |
| d) | What is the unique phenomenon that nonlinear systems exhibit? Discuss about the same. | 4 | L2 | 3 | 1 |

Unit – IV

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| 7. a) | What is Experimental Modal analysis? How does it help in modal analysis of a structural or mechanical system? | 8 | L2 | 4 | 1 |
| b) | Derive Parseval's formula for non-periodic functions. | 8 | L3 | 4 | 1 |
| c) | What are the characteristics of a response function? | 4 | L2 | 4 | 1 |
| 8. a) | Discuss different vibration monitoring techniques. | 7 | L2 | 4 | 1 |
| b) | Show that $R(\zeta) = R(-\zeta)$ | 5 | L3 | 4 | 1 |
| c) | Show that Frequency response function $H(\omega)$ is the Fourier Transform of the impulse response function. | 5 | L3 | 4 | 1 |
| d) | Discuss the significance of machine condition monitoring techniques. | 3 | L2 | 4 | 1 |

Unit – V

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|--------|---|----|----|---|---|
| 9. a) | Derive the equation of motion for longitudinal vibration of a rod and determine its solution with one end fixed and one end free. | 12 | L3 | 5 | 1 |
| b) | Discuss the orthogonality property of normal modes. | 8 | L2 | 5 | 1 |
| 10. a) | Determine the free vibration solution for lateral vibration of beams. | 12 | L3 | 5 | 1 |
| b) | What is the significance of Eigen value problem? Discuss a method to solve the same. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
